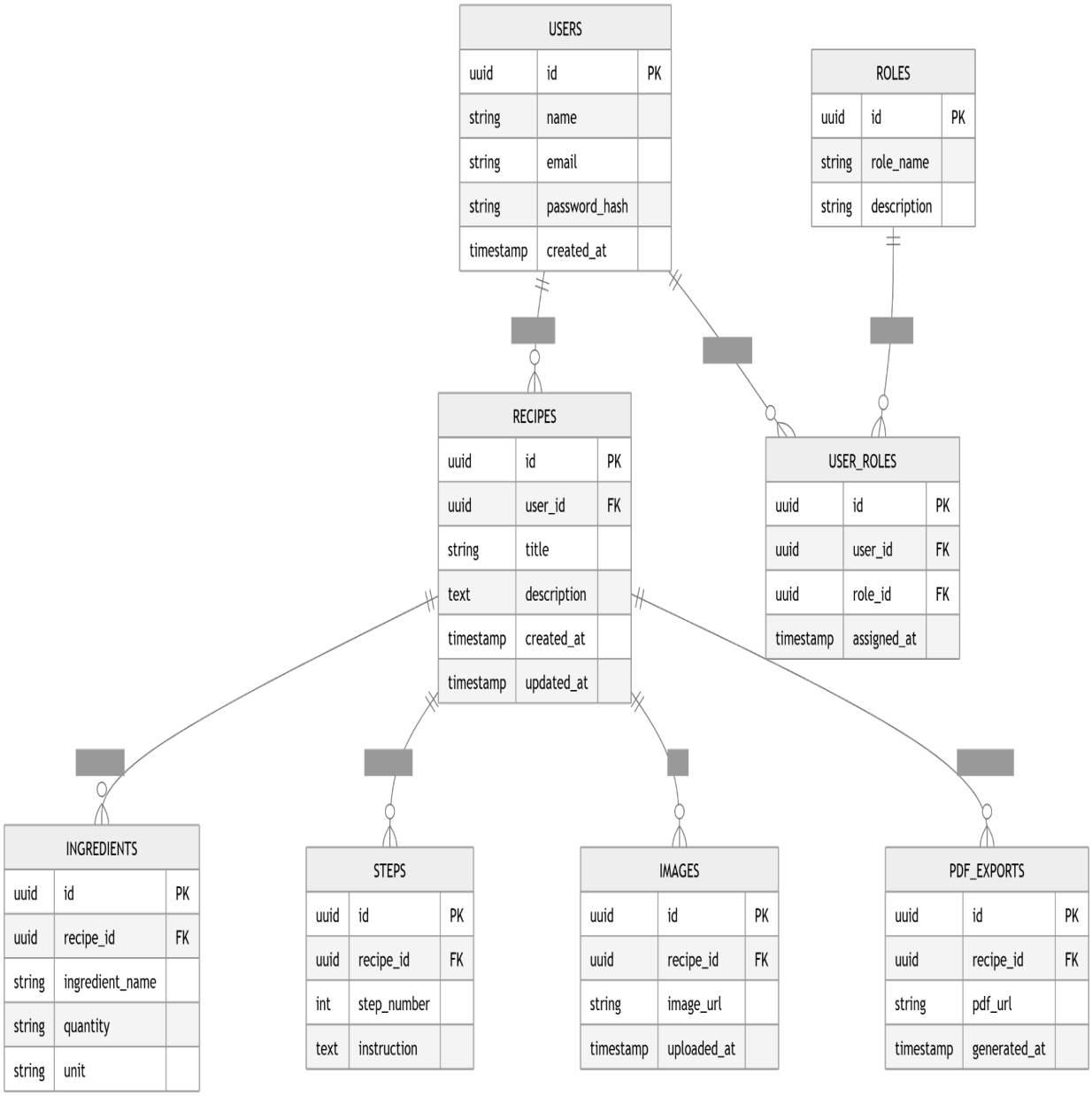
2. Database Schema Documentation

The system uses a relational database to manage users, recipes, ingredients, images, and role-based access.

Table	Purpose
USERS	Stores user accounts and authentication information
ROLES	Defines system roles such as admin or user
USER_ROLES	Maps users to roles
RECIPES	Stores recipe information
INGREDIENTS	Stores recipe ingredients
STEPS	Stores cooking instructions
IMAGES	Stores recipe images
PDF_EXPORTS	Stores generated recipe PDFs

Advanced ER Diagram


USERS
uuid id PK
string name
string email
string password_hash
timestamp created_at
ROLES
uuid id PK
string role_name
string description
RECIPES
uuid id PK
uuid user_id FK
string title
text description
timestamp created_at
timestamp updated_at
USER_ROLES
uuid id PK
uuid user_id FK
uuid role_id FK
timestamp assigned_at
INGREDIENTS
uuid id PK
uuid recipe_id FK
string ingredient_name
string quantity
string unit
STEPS
uuid id PK
uuid recipe_id FK
int step_number
text instruction
IMAGES
uuid id PK
uuid recipe_id FK
string image_url
timestamp uploaded_at
PDF_EXPORTS
uuid id PK
uuid recipe_id FK
string pdf_url
timestamp generated_at

Row Level Security (RLS)

Row Level Security (RLS) is a database-level security feature that ensures users can access only the specific rows (records) they are authorized to view or modify. It enforces access control directly within the database, providing strong and reliable data protection.

1. What is Row Level Security?

Row Level Security restricts access to individual rows in a database table. For example, if multiple interns create recipes in the same table: • Each intern can view and edit only their own recipes. • Mentors can view all recipes. • Public users can view only approved recipes. These rules are defined as policies inside the database.

2. Importance in Real-World Projects

In multi-user systems, strong access control is essential. Without RLS: • Users might access or modify data that does not belong to them. • Unapproved content could become publicly visible. • Sensitive internal information could be exposed. RLS prevents these risks by enforcing strict database-level policies.

3. Application in the Native Indian Recipes Project

In the Native Indian Recipes Internship Project, different user roles exist: Interns: • Create and manage their own recipes. • Edit or delete their own drafts only. Mentors/Reviewers: • Access all submitted recipes. • Approve or provide feedback. Public Users: • View only approved and published recipes. • Cannot edit or delete content. AI Documentation Records: • Accessible only to authorized users. RLS ensures proper separation of responsibilities and secure collaboration.

4. Technical Overview

In databases such as PostgreSQL or Supabase, RLS policies define rules like: • Allow SELECT only if user_id equals the logged-in user. • Allow UPDATE only if created_by matches the current user. These policies execute automatically whenever queries run, ensuring consistent enforcement of access rules.

5. Key Benefits

• Strong data protection • Role-based access control • Prevention of unauthorized modifications • Improved transparency in internship evaluation • Industry-standard security implementation Implementing RLS enhances the professionalism and reliability of the digital platform.

6. Conclusion

Row Level Security is a critical feature for building secure, multi-user platforms. For the Native Indian Recipes project, it ensures structured access control, secure collaboration, and controlled publishing of content. By implementing RLS, the platform aligns with real-world software engineering and security best practices.